

Junxiang (Sid) Gong

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EDUCATION

Cornell University

Master of Regional Planning, College of Architecture, Art & Planning

Coursework: Machine Learning and Artificial Intelligence; Urban Data Analytics; Systems Engineering and Optimization; Regional Economic Analysis.

Ithaca, NY

Aug 2025 – May 2027

Hainan University & Arizona State University

B.S. in Human Geography & Urban–Rural Planning, Haiya International College | GPA: 3.88/4.00, Rank: 10/103

Haikou, China

Sep 2021 – Jun 2025

Honors: First Prize, 2024 Esri GIS Development Competition (Geospatial Solution Design); Puyi Award (Bilingual Communication); IEERA (Applied Innovation); Dean's Scholarship (Academic Excellence); University Scholarships (Academic Merit); Outstanding Youth League Member (Leadership); Outstanding Volunteer (Collaboration).

PUBLICATION

Gong, J. (2024). "Strategic Analysis on Improving the Proximity and Accessibility in China: An Exploratory Development Based on the Study of the 15-Minute City." *Highlights in Science, Engineering and Technology*, 86, 66–73 (CEAT 2024).

EXPERIENCE

Digital Transformation Research Assistant

May 2026 – Present

China Academy of Information and Communications Technology (CAICT)

Shenzhen, China

- Audit 42 business modules and 1,134 function points for the *Shuzhiyi* enterprise digital-diagnostics platform; clean and catalog 164 industrial digital products across 27 categories; apply RAG, agent workflows, and tool calling to connect knowledge retrieval, product matching, and report generation, reducing report-processing time by approximately 30%.
- Lead 12 interviews across 6 manufacturers and translate business pain points into platform requirements. For a Shenzhen jewelry-cluster engagement, review 21 source files and an approximately 115-page acceptance report covering 2 firms, completing system diagnostics and evidence-chain validation.
- Develop enterprise digital-transformation consulting proposals; for a leading server vendor, analyze 55 field documents and drive approximately 20 revisions of a 58-page implementation plan through finalization and rollout.

Research Assistant, Prof. John I. Carruthers

Oct 2025 – Present

Cornell University

Ithaca, NY

- Build databases from BEA, BLS, FRED, Census, and GIS sources covering 3,080 U.S. counties and 99 indicators, plus a 5-year, 31-province Chinese panel with 2,790 observations and 45 variables; standardize identifiers, definitions, and time ranges for merging and spatial matching.
- Apply location quotients, instrumental-variable models, panel random effects, and gravity models in Stata, Python, QGIS, and Excel to compare industrial concentration and migration; maintain datasets and results in active use by the research group.
- Build a 67-sector 2022 U.S. input–output model, trace a hypothetical construction-demand shock through industry linkages, and estimate an output multiplier of approximately 1.76; deliver a 10-page English report, 5 economic exhibits, and GIS maps for faculty review.

Strategic Research Intern

Feb 2025 – Jun 2025

ZTE Corporation

Beijing, China

- Synthesized 40 white papers, policy documents, and market sources on advanced displays, semiconductors, and biopharma; assessed capacity, market size, value chains, and competition to identify priority segments, growth opportunities, and entry risks.
- Surveyed 7 industrial parks in 2 weeks and compared transport, industrial support, schedules, and operating costs across 5 candidate plans; converted findings into siting logic, industry positioning, and a phased roadmap.
- Contributed to more than 20 planning and research deliverables; 3 siting, positioning, and implementation recommendations entered the final plan. Prepared client presentations and review materials supporting successful preliminary expert review.

SELECTED PROJECTS

AI and Spatial ML for Multimodal Transit Coverage and Area Segmentation | Project Lead

Feb 2026 – Jun 2026

- Integrated ACS, TIGER/Line, and OpenStreetMap data into a unified geospatial database covering 303 census tracts and 3,221 bus, subway, and bike-share service nodes; framed resident accessibility as a demand-area-to-service-node matching problem.
- Built a pedestrian network with OSMnx, calculated nearest-service walking times using multi-source Dijkstra, and standardized features with RobustScaler; applied K-means ($k = 4$) to identify high-coverage Lower/Midtown cores, low-coverage waterfront areas, and subway-deficit zones. OLS results link a 10-percentage-point increase in poverty share to 0.33- and 0.54-minute longer minimum walks to bus and subway service, informing gap-filling recommendations for NYC transportation planners.

Stay or Return Decision-Support Platform | Creator | stayorreturn.com

Nov 2025 – Present

- Structure the U.S.–China decision across 6 dimensions; design 24 questions, 72 options, and a 4-stage assessment, weighting, explanation, and decision-memo flow. Develop preference-scoring and sensitivity-analysis methods that expose dimension scores, key drivers, and recommendation logic.
- Build questionnaire, scoring, visualization, and sharing features with Next.js, TypeScript, and AI-assisted coding tools; deploy on Vercel. Test scoring logic, boundary inputs, and critical paths; incorporate feedback to refine information architecture and results presentation.

Truck–Drone Fulfillment Network and Collaborative Dispatch Optimization | Project Lead

Dec 2024 – Jun 2025

- Modeled a 2.37 km², 50-node truck–drone system as a mixed-integer program with 17 constraint classes; engineered demand, depot-distance, node-density, and road-detour features, then combined Gaussian-mixture clustering, nearest-neighbor/TSP initialization, genetic-algorithm search, and Gurobi optimization to reach a 1.6% MIP gap within 1,800 seconds.
- Against a vehicle-only CVRP baseline, reduced total cost by 5.2%, average completion time by 36.4%, and the weighted objective by 6.9% in the 50-node test; documented methods, results, and dispatch recommendations in an English report and reproducible codebase.

SKILLS

Research & Analytics: Python, scikit-learn, OSMnx, SQL, Excel, Stata, QGIS, and Gurobi for data cleaning, metric design, econometric analysis, spatial analysis, visualization, and optimization.

AI Applications: RAG, agent workflows, tool calling, prompt/context design, source-grounded generation, and baseline evaluation.

Product & Solutions: B2B customer interviews, process mapping, requirements analysis, PRDs, prototyping, and front-/back-end coordination.

Web Development: Next.js, TypeScript, web prototyping, debugging, and Vercel deployment. **Languages:** English (professional proficiency); Mandarin Chinese (native).